

## Ch. 3. Chemical naming

**A**LL elements in the periodic table except for the noble gases—He, Ne, Ar, Kr, Xe, and Rn—combine to produce chemical compounds. Most of these chemicals are useful in your everyday life, and you drink water to quench your thirst, use Clorox and ammonia to clean your house, or baking soda to deal with a stinky refrigerator. In this chapter you will learn not only how to name these chemicals but also how to read chemical formulas—we call this to formulate chemicals. Still, chemical elements such as hydrogen and oxygen do not combine randomly and they only choose specific elemental partners to form a compound. As an example, hydrogen combines with oxygen using specific proportions to produce  $\text{H}_2\text{O}$  and not  $\text{HO}_2$ . In this chapter, you will also learn the rules that chemical elements use to combine in special ratios.

### 3.1 Ions and ionic charges

Atoms gain and lose electron density to produce ions. An ion is just an atom with a positive or negative charge. Ions result from an electron density transfer. Positive ions have lost negatively charged electron density, whereas negative ions have gained electron density. The reason for this electronic transfer is that atoms try to achieve a very stable electronic configuration with eight electrons in the valence, and this is called the octet electron configuration. Examples of ions are:  $\text{H}^+$ ,  $\text{Ca}^{2+}$  or  $\text{O}^{2-}$ . This section covers the properties of ions and ionic charges.

*Cations and anions* Atoms that lose electron density become positively charged. These ions are called cations. Examples of cations are  $\text{Li}^+$  or  $\text{Mg}^{2+}$  called lithium cation and magnesium cation, respectively. Atoms that gain electron density become negatively charged, as electron density have a negative charge. These ions are called anions. Examples of anions are  $\text{F}^-$  called fluoride or  $\text{N}^{3-}$  called nitride. The way to name anions is by using the name of the element and the suffix -ide.

*Ionic charges: the valences* How do we know that hydrogen produces a  $\text{H}^+$  ion and nitrogen a  $\text{N}^{3-}$  anion? The charge of an ion is called an ionic charge, and the numbers are coming from the periodic table. H, Na, or K are in group IA (left of the table) and hence the ionic charge will be  $1+$ . Similarly, Mg or Ca are in group IIA (left of the table) and hence the ionic charge will be  $2+$ . Differently, F, Cl, or Br are in group 7A (right of the table) and their charge will be  $1-$ . Oxygen is in group 6A (right of the table) and the ionic charge will be  $2-$ . Figure ?? contains all ionic charges. What if the element is not on this list such as in the case of Iron (Fe)? In that case, very probably it will have several ionic charges and



this charge has to be indicated in the chemical name. An example would be Fe, which ionic charge is not in Figure ?? as iron can have several ionic charges.

| 1  | 2               | 3                | 4 | 5 | 6 | 7                                    | 8                                    | 9                                    | 10                                   | 11 | 12                                  | 13               | 14                                   | 15                                   | 16              | 17              | 18 |
|----|-----------------|------------------|---|---|---|--------------------------------------|--------------------------------------|--------------------------------------|--------------------------------------|----|-------------------------------------|------------------|--------------------------------------|--------------------------------------|-----------------|-----------------|----|
| 1A | 2A              |                  |   |   |   |                                      |                                      |                                      |                                      |    |                                     | 3A               | 4A                                   | 5A                                   | 6A              | 7A              | 8A |
| 1  | H <sup>+</sup>  |                  |   |   |   |                                      |                                      |                                      |                                      |    |                                     | B <sup>3+</sup>  |                                      | N <sup>3-</sup>                      | O <sup>2-</sup> | F <sup>-</sup>  |    |
| 2  | Li <sup>+</sup> | Be <sup>2+</sup> |   |   |   |                                      |                                      |                                      |                                      |    |                                     | Al <sup>3+</sup> |                                      | P <sup>3-</sup>                      | S <sup>2-</sup> | Cl <sup>-</sup> |    |
| 3  | Na <sup>+</sup> | Mg <sup>2+</sup> |   |   |   |                                      |                                      |                                      |                                      |    |                                     |                  |                                      |                                      |                 |                 |    |
| 4  | K <sup>+</sup>  | Ca <sup>2+</sup> |   |   |   | Cr <sup>2+</sup><br>Cr <sup>3+</sup> | Mn <sup>2+</sup><br>Mn <sup>3+</sup> | Fe <sup>2+</sup><br>Fe <sup>3+</sup> | Co <sup>2+</sup><br>Co <sup>3+</sup> |    | Cu <sup>+</sup><br>Cu <sup>2+</sup> | Zn <sup>2+</sup> | Ga <sup>3+</sup>                     |                                      |                 | Br <sup>-</sup> |    |
| 5  | Rb <sup>+</sup> | Sr <sup>2+</sup> |   |   |   |                                      |                                      |                                      |                                      |    | Ag <sup>+</sup>                     | Cd <sup>2+</sup> |                                      | Sn <sup>2+</sup><br>Sn <sup>4+</sup> |                 | I <sup>-</sup>  |    |
| 6  | Cs <sup>+</sup> | Ba <sup>2+</sup> |   |   |   |                                      |                                      |                                      |                                      |    |                                     |                  | Pb <sup>2+</sup><br>Pb <sup>4+</sup> |                                      |                 |                 |    |

Figure ?? Ionic charges (valences) for different elements

### Sample Problem 1

Identify the correct ionic state of the following elements: (a) Cl (b) K (c) O (d) C

#### SOLUTION

Cl is on the 7A group and hence its charge is 1−, whereas potassium belongs to 1A and its charge will be 1+. Oxygen and carbon will have 2− and 4− charges. The final ionic states are: Cl<sup>−</sup>, K<sup>+</sup>, O<sup>2−</sup> and C<sup>4−</sup>.

#### STUDY CHECK

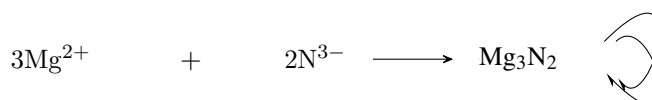
Identify the correct ionic state of the following elements: (a) N (b) Br

►Answer: (a) N<sup>3−</sup> (b) Br<sup>−</sup>

## 3.2 Ionic compounds

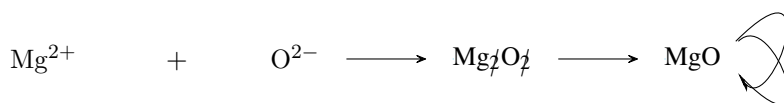
Ionic compounds result from the combination of a nonmetallic element with a metallic element. An example is NaCl, which results from combining sodium (a metal) with chloride (a nonmetal). Ionic compounds normally have high melting points and are solid under normal conditions. A typical ionic compound would be NaCl, cooking salt. Atoms of an ionic compound are connected through an ionic bond. In an ionic bond, one element gives away (the cation) and the other one receives electron density (the anion). As an example, in the NaCl molecule Na gives away electron density to Cl, and the molecule results from the combinations of Na<sup>+</sup> and Cl<sup>−</sup>. In an ionic compound, the element on the left is positive and the one on the right is negative.

*Combining ions* Ionic compounds are the result of combining two ions: a positive (cation) and a negative (anion) ion. Each ion has a charge, depending on its location on the periodic table. When combining two atoms you first need to arrange the ions starting from positive and followed by negative. The charges of an ion would become the coefficient of the other ion. For example Mg<sup>2+</sup> and N<sup>3−</sup> are combined as Mg<sub>3</sub>N<sub>2</sub>:





Another example would be the combination of  $\text{Na}^+$  and  $\text{O}^{2-}$  that would be  $\text{Na}_2\text{O}$ . You need to simplify the indexes of the formula by dividing by the smallest one, always using integer values. For example,  $\text{Mg}^{2+}$  and  $\text{O}^{2-}$  combine to give  $\text{Mg}_2\text{O}_2$  that should be written as  $\text{MgO}$



Another naming example that involves simplifying the formula is the chemical result of combining  $\text{Ca}^{2+}$  and  $\text{C}^{4-}$ . After combining the charges we obtain  $\text{Ca}_4\text{C}_2$  that needs to be simplified by dividing by the smallest number leading to  $\text{Ca}_2\text{C}$ .

#### Sample Problem 2

Combine the following ions or give the ions given the final compound: (a)  $\text{Li}^+$  and  $\text{O}^{2-}$  (b)  $\text{Ca}^{2+}$  and  $\text{O}^{2-}$  (c)  $\text{Li}_3\text{N}$  (d)  $\text{Mg}_2\text{C}$

#### SOLUTION

The result of combining  $\text{Li}^+$  and  $\text{O}^{2-}$  is  $\text{Li}_2\text{O}$ . For  $\text{Ca}^{2+}$  and  $\text{O}^{2-}$ , the resulting chemical is  $\text{CaO}$ .  $\text{Li}_3\text{N}$  results from the combination of  $\text{Li}^+$  and  $\text{N}^{3-}$ , and  $\text{Mg}_2\text{C}$  results from  $\text{Mg}^{2+}$  and  $\text{C}^{4-}$ .

#### STUDY CHECK

Combine the following ions or give the ions given the final compound: (a)  $\text{Na}^+$  and  $\text{F}^-$  (b)  $\text{Na}_3\text{N}$

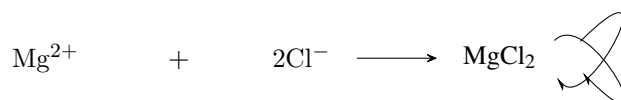
►Answer: (a)  $\text{NaF}$  (b)  $\text{Na}^+$  and  $\text{N}^{3-}$

*Simple ionic naming (type I ionic)* Type I ionic compounds result from the combination of a metal with given valence (Li, Ca, Mg, etc, see Figure ?? ) and a nonmetal. To name an ionic compound (type I ionic) you need to (a) use the name of the first element in the compound, (b) use the first syllable of the second element, and (c) finish the name of the molecule in the suffix *-ide*. As an example, the formula  $\text{NaCl}$  is named sodium chloride, and  $\text{MgCl}_2$  is named magnesium chloride. Another example would be:

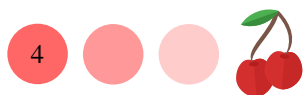
calcium chloride

$\text{CaCl}_2$  (ionic)

To formulate an ionic compound from a given name, we need to combine both ions by exchanging the valences (the ionic charges). For example,  $\text{MgCl}_2$  results from the combination of  $\text{Mg}^{2+}$  and  $\text{Cl}^-$  so that the number 2 in  $\text{MgCl}_2$  near the Cl atom is coming from the  $\text{Mg}^{2+}$ . In other words:



The sign of the charges only indicates which element goes first in the formula: the positive element (cation) first followed by the negative element (anion). For example, the result of combining  $\text{Na}^+$  and  $\text{Cl}^-$  is  $\text{NaCl}$  and not  $\text{ClNa}$  as Na has a positive ionic charge and has to appear first in the formula.



## Sample Problem 3

Name or formulate the following ionic compounds: (a)  $\text{MgO}$  (b)  $\text{Mg}_3\text{N}_2$  (c) Lithium nitride (d) Magnesium carbide

**SOLUTION**

The name for  $\text{MgO}$  is magnesium oxide.  $\text{Mg}_3\text{N}_2$  is called magnesium nitride. The formula for Lithium nitride is  $\text{Li}_3\text{N}$  and the formula for Magnesium carbide is  $\text{Mg}_2\text{C}$ , result of simplifying  $\text{Mg}_4\text{C}_2$  dividing by two, the smallest number.

**STUDY CHECK**

Name or formulate the following ionic compounds: (a) Sodium fluoride (b)  $\text{Na}_3\text{N}$

►Answer: (a)  $\text{NaF}$  (b) Sodium nitride

$\text{NaCl}$  results from the combination of  $\text{Na}^+$  and  $\text{Cl}^-$ . The ionic charges of  $\text{Na}$  and  $\text{Cl}$  are given in Figure ?? according to the group. If the ionic chemical contains a transition metal with variable ionic charge, that is, which is not in Figure ?? then the ionic naming becomes a bit more complex. The reason is that one needs to specify the charge of the metal, explicitly in the name of the chemical. An example would be  $\text{NiCl}_2$  named as Nickel(II) chloride or  $\text{Co}_2\text{O}_3$  named as Cobalt(III) oxide.

*Name complex ionic chemicals* This section covers how to name ionic chemicals containing a metal with variable charge. In this case, you need to specify the charge of the metal in the name. To calculate this number you will solve a simple math equation. For example, the name of  $\text{Mn}_2\text{O}_3$  is Manganese(III) oxide. How do we get this name? Manganese has several charges as it is not in Figure ??, let us use  $x$  for its charge  $\text{Mn}^x$  and oxygen has a charge of two  $\text{O}^{2-}$ . After combining  $\text{Mn}^x$  and  $\text{O}^{2-}$  the resulting formula would be  $\text{Mn}_2\text{O}_x$ . By comparison with the given formula,  $\text{Mn}_2\text{O}_3$ ,  $x$  has to be three and hence the charge of  $\text{Mn}$  has to be three. Therefore, the final name would be Manganese(III) oxide.

## Sample Problem 4

Name or formulate the following ionic compounds: (a)  $\text{MnO}$  (b)  $\text{Fe}_3\text{N}_2$  (c) Cobalt(II) carbide (d) Iron(II) oxide

**SOLUTION**

All the chemicals on this example contain a metal that can have several charges, and hence, we need to specify the ionic charge on the name.  $\text{MnO}$  results from  $\text{Mn}^x$  and  $\text{O}^{2-}$ . After combining the ions, the formula would be  $\text{Mn}_2\text{O}_x$ , a formula that needs to be compared to  $\text{MnO}$ . The formulas do not look similar, so let's make them more similar by dividing by two so that  $\text{MnO}_{\frac{x}{2}}$  resembles  $\text{MnO}$ . By comparing  $x$  has to be 2 and hence the name is Manganese(II) oxide. The name for  $\text{Fe}_3\text{N}_2$  would be Iron(II) nitride. The valence of Iron comes from combining  $\text{Fe}^x$  and  $\text{N}^{3-}$  that gives  $\text{Fe}_3\text{N}_x$ . By comparison with  $\text{Fe}_3\text{N}_2$   $x$  has to be two and the name is Iron(II) nitride. the formula for Cobalt(II) carbide would be  $\text{Co}_2\text{C}$  as Cobalt(II) is  $\text{Co}^{2+}$  and carbide is  $\text{C}^{4-}$ . After combining the ions one obtains  $\text{Co}_4\text{C}_2$  that gives  $\text{Co}_2\text{C}$ . Finally, the formula for Iron(II) oxide is  $\text{FeO}$  as Iron(II) is  $\text{Fe}^{2+}$  and oxide is  $\text{O}^{2-}$  that gives  $\text{Fe}_2\text{O}_2$  and simplifying one obtains  $\text{FeO}$ .

**STUDY CHECK**

Name or give the formula for the following ionic compounds: (a) Manganese(IV)

oxide (b)  $\text{AuCl}_3$ ►Answer: (a)  $\text{MnO}_2$  (b) Gold(III) chloride

### 3.3 Covalent compounds

Covalent compounds are chemicals resulting from the combination of nonmetallic elements. An example is  $\text{CO}_2$ , which results from combining carbon (a nonmetal) with oxygen (a nonmetal). At normal conditions, covalent compounds may exist as solids, liquids, or gases. Covalent compounds do not exhibit any electrical conductivity, either in pure form or when dissolved in water. A typical covalent compound would be  $\text{H}_2\text{O}$ , water. Atoms in a covalent compound are connected by means of a covalent chemical bond. In a covalent bond, both atoms connected share the electrons. As an example, the  $\text{HCl}$  molecule has an hydrogen and a chlorine atom connected by means of a covalent bond, in which each atom shares the electrons of the bond.

**Covalent naming** To name a covalent compound you need to (a) use the name of the first element in the compound, (b) use the first syllable of the second element, and (c) finish the name of the molecule in the suffix *-ide*. More importantly, you need to use prefixes (see Table ??) that indicate the number of atoms in the molecule. See the Table ?? below for a list of the different equivalencies between prefixes and numbers. As an example, the formula  $\text{CH}_4$  is named carbon tetrahydride. Similarly, a covalent chemical name can be translated into a chemical formula (we call this to formulate a chemical with a given name), and the formula for carbon monoxide would be  $\text{CO}$ . Another example would be  $\text{N}_2\text{O}$  named as dinitrogen oxide, and the name sulfur hexafluoride corresponds to the formula  $\text{SF}_6$ . The prefix *mono* is omitted in the first element of the name, and for example, you will not name the chemical  $\text{CO}$  as ~~monocarbon~~ monoxide, you would just say carbon monoxide. When the vowels *a* and *o* appear together, the first vowel is omitted as in pentoxide instead of ~~pentaoxide~~. A final example of a covalent compound would be:

dinitrogen pentoxide

 $\text{N}_2\text{O}_5$  (covalent)

**Table ?? Prefixes used to name covalent compounds**

| Prefix | number | Prefix | number | Prefix | number |
|--------|--------|--------|--------|--------|--------|
| Mono   | 1      | Tetra  | 4      | Hepta  | 7      |
| Di     | 2      | Penta  | 5      | Octa   | 8      |
| Tri    | 3      | Hexa   | 6      | Nona   | 9      |

#### Sample Problem 5

Name or formulate the following covalent chemicals: (a)  $\text{NO}$  (b)  $\text{CS}_2$  (c) Sulfur Dioxide (d) Nitrogen Trichloride

#### **SOLUTION**

All chemicals in this example are covalent as they result of the combination of nonmetals. In order to name them, we need to use prefixes and finish the suffix



with -ide. The first chemical is called nitrogen monoxide.  $\text{CS}_2$  is called carbon disulfide. The formula for sulfur dioxide and nitrogen trichloride are respectively  $\text{SO}_2$  and  $\text{NCl}_3$ .

### STUDY CHECK

Name or formulate the following covalent chemicals: (a)  $\text{SCl}_2$  (b) diboron trioxide

►Answer: (a) sulfur dichloride (b)  $\text{B}_2\text{O}_3$

## 3.4 Naming acids & bases

In this section, we will learn how to name acids and bases. Acids normally have common names (e.g. sulfuric acid) and their naming does not follow modern rules. Names and formulas of acids are commonly tabulated due to historical reasons. Differently, bases (e.g. sodium hydroxide) are named in a standard way.

**Bases or hydroxides** Bases (hydroxides) result from the combination of metal and the hydroxide anion ( $\text{OH}^-$ ). Examples are  $\text{NaOH}$  or  $\text{Ca}(\text{OH})_2$ . The name of a base starts with the name of the cation finishing with the word *hydroxide*. An example is  $\text{NaOH}$  named as *sodium hydroxide*, or  $\text{Ca}(\text{OH})_2$ , named as *calcium hydroxide*. The word *hydroxide* refers to the  $\text{OH}^-$  ion, and hence Sodium hydroxide results from combining  $\text{Na}^+$  and  $\text{OH}^-$ , and Calcium hydroxide from combining  $\text{Ca}^{2+}$  and  $\text{OH}^-$ . More examples of hydroxides:

Magnesium hydroxide

$\text{Mg}(\text{OH})_2$ (hydroxide)

As a final note, not all bases are hydroxides. For example, ammonia  $\text{NH}_3$  is a base even when it does not contain hydroxides in its structure. A way to remember this is sometimes to write ammonia like  $\text{NH}_4\text{OH}$ .

**Acids** Acids—in particular inorganic acids—are chemicals that normally contain hydrogen at the beginning of their formula. For example,  $\text{HCl}$  or  $\text{H}_2\text{SO}_4$ .  $\text{HCl}$  is a hydric acid and is named as *hydrochloric acid*, whereas  $\text{H}_2\text{SO}_4$  is an oxoacid that contains oxygen named as *sulfuric acid*. The names of acids are not standard and they come from common names employed in the field for many years. Table ?? contains a list of the most important oxoacids and hydric acids. More examples of acids:

Nitric acid

$\text{HNO}_3$ (oxoacid)

Hydrofluoric acid

$\text{HF}$ (hydric acid)

### Sample Problem 6

Name or formulate the following acids and bases. Indicate whether the compound is an acid or a base. (a)  $\text{HCN}$  (b)  $\text{KOH}$  (c) Carbonic acid (d) Lithium hydroxide

### SOLUTION

$\text{HCN}$  is an acid named hydrocyanic acid.  $\text{KOH}$  is a base called potassium



hydroxide. The formula for Carbonic acid is  $\text{H}_2\text{CO}_3$ , and Lithium hydroxide is a base with formula  $\text{LiOH}$ .

### ◆ STUDY CHECK

Name or formulate the following ionic compounds: (a) phosphoric acid  
(b)  $\text{Mg}(\text{OH})_2$

►Answer: (a)  $\text{H}_3\text{PO}_4$  (b) magnesium hydroxide

## 3.5 Oxidation states

Isolated atoms tend to have a neutral charge. However, the atoms of elements have the capacity of gaining and losing electron density forming cations and anions. The atoms that form a compound can have different charge states resulting in losing and gaining electronic charge. We refer to this as the oxidation state of an element in a compound.

*Oxidation states of oxoacids* Consider the following set of acids:  $\text{HClO}$ ,  $\text{HClO}_2$ ,  $\text{HClO}_3$  and  $\text{HClO}_4$ . We say Cl in these acids have different oxidation states or different oxidation numbers. This section will cover the calculation of the oxidation state of the central atom of an oxoacid.

Let us address the oxoacid:  $\text{H}\underline{\text{Cl}}\text{O}_3$ . The goal is to calculate the oxidation number of the underlined element, Cl. To do this we will follow a set of simple rules. First, we will use the valences as the oxidation number of the elements (see Figure ??) to the right and the left of the central atom. Then, we will assign an unknown oxidation state of  $x$  to the central atom. After that, we will set up an equation so that the sum of all oxidation numbers equals the charge of the acid if any. In this formula, we will include the atomic coefficients. In the case of  $\text{H}\underline{\text{Cl}}\text{O}_3$ , the equation would be:

$$1 + x + 3 \cdot (-2) = 0$$

as the number of oxygens is three, we will have to time by three the valence of oxygen. The number zero results from the charge of the acid. If we solve for  $x$ , we obtain  $x = 5$ . That is, the oxidation state of Cl on  $\text{H}\underline{\text{Cl}}\text{O}_3$  is 5 and this is represented as  $\text{H}\underline{\text{Cl}}^{\text{V}}\text{O}_3$ .

*Oxidizing and reducing character of oxoacids* The importance of the oxidation state of the central elements of oxoacid results from the fact that acids with high or low oxidation states, tend to be very reactive, sometimes capable of completely dissolving metals. We call this oxidizing (or reducing) acids. For example,  $\text{HNO}_3$  and  $\text{H}_2\text{SO}_4$  are both oxidizing acids, and these acids will solve for example a piece of copper. Similarly, acids with very small or negative oxidation numbers can be very reactive as well. These acids are called reducing acids or agents. Let us compare two oxoacids to elaborate more on the terminology used to describe redox numbers. For example, let us compare  $\text{H}\underline{\text{Cl}}^{\text{V}}\text{O}_3$  and  $\text{H}\underline{\text{Cl}}^{\text{III}}\text{O}_2$ . We say Cl on  $\text{H}\underline{\text{Cl}}^{\text{V}}\text{O}_3$  has a larger redox number than  $\text{H}\underline{\text{Cl}}^{\text{III}}\text{O}_2$ . We can also say, Cl in  $\text{H}\underline{\text{Cl}}^{\text{V}}\text{O}_3$  is more oxidized than Cl on  $\text{H}\underline{\text{Cl}}^{\text{III}}\text{O}_2$ . Finally, we can also say,  $\text{H}\underline{\text{Cl}}^{\text{V}}\text{O}_3$  is more reducing than  $\text{H}\underline{\text{Cl}}^{\text{III}}\text{O}_2$ . Again, the terms associated with high redox numbers are oxidized and reducing, and the terms associated with low redox numbers are reduced and oxidizing.

It is important to note that the oxidation state of an element is ultimately related to the amount of electron density—electronic charge—of that element. The more electronic charge the smaller—the more negative—the oxidation state. In other words, large oxidation states result from losing electron density.



## Sample Problem 7

Calculate the redox number of S in the following acids and indicate the more oxidizing acid:  $\text{H}_2\text{S}_2\text{O}_6$  named dithionic acid and  $\text{H}_2\text{SO}_4$  named sulfuric acid.

**SOLUTION**

We will set up the redox formula for the first acid ( $\text{H}_2\text{S}_2\text{O}_6$ ), given that the redox number of H is +1 and the redox number of O is -2.

$$2 \cdot 1 + 2 \cdot x + 6 \cdot (-2) = 0$$

Solving for x:

$$2 + 2 \cdot x - 12 = 0 \quad \text{we have that } x = \frac{12 - 2}{2}$$

The oxidation state of S in  $\text{H}_2\text{S}_2\text{O}_6$  is +5. For the second acid ( $\text{H}_2\text{SO}_4$ ):

$$2 \cdot 1 + x + 4 \cdot (-2) = 0$$

Solving for x:

$$2 + x - 8 = 0 \quad \text{we have that } x = \frac{8 - 2}{1}$$

that gives a redox of 6. If we compare both acids the smaller the redox number the more reduced is the central element and the more oxidizing the acid is. Therefore,  $\text{H}_2\text{S}_2\text{O}_6$  is more oxidizing than  $\text{H}_2\text{SO}_4$ .

**STUDY CHECK**

Calculate the redox number of the following acids: (a)  $\text{H}_2\text{MnO}_4$  (b)  $\text{H}_2\text{Cr}_2\text{O}_7$

►Answer: (a) +6 (b) +6

## 3.6 Naming complex salts & common chemicals

At this point, we saw the naming and formulation of ionic (e.g. NaCl) and covalent compounds (e.g.  $\text{CO}_2$ ). This section covers the naming of complex salts: oxosalts and hydrosalts. In general, salts (oxosalts or hydrosalts) are the result of mixing an oxoacid and a base. Their formulas tend to look more complex than simple ionic or covalent compounds as they are made of at least three different elements. An example of an oxosalt would be  $\text{CaSO}_4$  called calcium carbonate. An example of hydrosalt would be  $\text{NaHSO}_4$  which is called sodium monohydrosulfate. This section will also cover the naming of hydrates (e.g.  $\text{CaSO}_4 \cdot \text{H}_2\text{O}$ ), which are compounds containing water molecules inside their structure. Before being able to name these complex chemicals it is convenient to practice combining ions, without paying attention to the chemical naming.

*Combining ions* To combine two ions, you first arrange the positive ion on the left followed by the negative ion on the right, to then cross the ionic charges from the top of the ion to the bottom of the opposite ion. The positive and negative charges are not carried. If the ions have more than one element we have to use parenthesis. An example would be combining  $\text{Ca}^{2+}$  and  $\text{PO}_4^{3-}$  leading to  $\text{Ca}_3(\text{PO}_4)_2$ :







We would need to simplify the formula in case the charges compensate each other.

An example would be combining  $\text{Mg}^{2+}$  and  $\text{SO}_4^{2-}$  leading to  $\text{MgSO}_4$

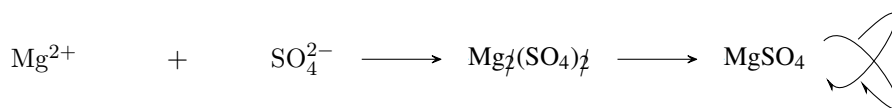


Table ?? Names of oxoacids and oxoalts (top table) and hydracids (bottom table).\*

| Element    | Oxoacid                           | Oxoacid Name          | Oxoasalt                     | Oxoasalt Name   |
|------------|-----------------------------------|-----------------------|------------------------------|-----------------|
| Manganese  | $\text{HMnO}_4$                   | Permanganic Acid      | $\text{MnO}_4^-$             | Permanganate    |
|            | $\text{H}_2\text{MnO}_4$          | Manganic acid         | $\text{MnO}_4^{2-}$          | Manganate       |
| Carbon     | $\text{H}_2\text{CO}_3$           | Carbonic Acid         | $\text{CO}_3^{2-}$           | Carbonate       |
| Nitrogen   | $\text{HNO}_3$                    | Nitric Acid           | $\text{NO}_3^-$              | Nitrate         |
|            | $\text{HNO}_2$                    | Nitrous Acid          | $\text{NO}_2^-$              | Nitrite         |
| Phosphorus | $\text{H}_3\text{PO}_4$           | Phosphoric Acid       | $\text{PO}_4^{3-}$           | Phosphate       |
| Sulfur     | $\text{H}_2\text{SO}_4$           | Sulfuric Acid         | $\text{SO}_4^{2-}$           | Sulfate         |
|            | $\text{H}_2\text{SO}_3$           | Sulfurous Acid        | $\text{SO}_3^{2-}$           | Sulfite         |
|            | $\text{H}_2\text{S}_2\text{O}_2$  | Thiosulfurous Acid    | $\text{S}_2\text{O}_2^{2-}$  | Thiosulfite     |
|            | $\text{H}_2\text{S}_2\text{O}_3$  | Thiosulfuric Acid     | $\text{S}_2\text{O}_3^{2-}$  | Thiosulfate     |
|            | $\text{H}_2\text{S}_2\text{O}_7$  | Disulfuric acid       | $\text{S}_2\text{O}_7^{2-}$  | Disulfate       |
|            | $\text{H}_2\text{S}_2\text{O}_8$  | Peroxodisulfuric acid | $\text{S}_2\text{O}_8^{2-}$  | Peroxodisulfate |
|            | $\text{HClO}_4$                   | Perchloric Acid       | $\text{ClO}_4^-$             | Perchlorate     |
|            | $\text{HClO}_3$                   | Chloric acid          | $\text{ClO}_3^-$             | Chlorate        |
| Chlorine   | $\text{HClO}_2$                   | Chlorous acid         | $\text{ClO}_2^-$             | Chlorite        |
|            | $\text{HClO}$                     | Hypochlorous acid     | $\text{ClO}^-$               | Hypochlorite    |
| Iodine     | $\text{HIO}_4$                    | Periodic Acid         | $\text{IO}_4^-$              | Periodate       |
| Chromium   | $\text{H}_2\text{CrO}_4$          | Chromic acid          | $\text{CrO}_4^{2-}$          | Chromate        |
|            | $\text{H}_2\text{Cr}_2\text{O}_7$ | Dichromic acid        | $\text{Cr}_2\text{O}_7^{2-}$ | Dichromate      |
| Boron      | $\text{H}_3\text{BO}_3$           | Boric acid            | $\text{BO}_3^{3-}$           | Borate          |
| Hydracid   | Hydracid Name                     | Hydracid              | Hydracid Name                |                 |
| HCl        | Hydrochloric acid                 | HBr                   | Hydrobromic acid             |                 |
| HI         | Hydroiodic acid                   | HF                    | Hydrofluoric acid            |                 |
| HCN        | Hydrocyanic acid                  | $\text{H}_2\text{S}$  | Hydrosulfuric acid           |                 |

\* Yellow indicate very important acids

**Naming Oxoalts** The names of the oxoalts are constructed by combining the name of the first element—you need to specify its charge in the case of a transition metal element with different possible charges—followed by the name of the oxoalt from Table ???. For example, the name of  $\text{MgSO}_4$  is magnesium sulfate, as  $\text{Mg}^{2+}$  is magnesium and  $\text{SO}_4^{2-}$  is sulfate. Another example is  $\text{Fe}_2(\text{CO}_3)_3$  called Iron(III) carbonate. A final example:

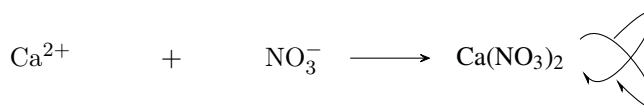
Lithium nitrate

$\text{LiNO}_3$ (oxoalt)

**Formulating Oxoalts** When you know the name of an oxoalt and you want to obtain its formula, you first need to arrange the positive ion on the left followed by the



negative ion on the right, to then cross the ionic charges from the top of the ion to the bottom of the opposite ion. For example, calcium nitrate results from the combination of  $\text{Ca}^{2+}$  calcium and  $\text{NO}_3^-$ , nitrate. By combining the two ions we obtain the final formula as  $\text{Ca}(\text{NO}_3)_2$ :



#### Sample Problem 8

Name or formulate the following oxosalts: (a)  $\text{K}_2\text{SO}_4$  (b)  $\text{Na}_2\text{CO}_3$  (c) Magnesium carbonate (d) Sodium phosphate

#### SOLUTION

$\text{K}_2\text{SO}_4$  is named potassium sulfate, as  $\text{K}^+$  is potassium and  $\text{SO}_4^{2-}$  stands for sulfate.  $\text{Na}_2\text{CO}_3$  is sodium carbonate. Magnesium carbonate is  $\text{MgCO}_3$  and sodium phosphate is  $\text{Na}_3\text{PO}_4$ .

#### STUDY CHECK

Name or formulate the following oxosalts: (a)  $\text{CaSO}_4$  (b) Aluminum sulfate

►Answer: (a) Calcium sulfate (b)  $\text{Al}_2(\text{SO}_4)_3$

*Naming oxosalts with hydrogen* Some oxosalts contain hydrogen atoms in their structure between the oxosalt cation and anion (e.g.  $\text{NaHSO}_4$ ). For example,  $\text{NaHSO}_4$  is named sodium monohydrogensulfate. To understand this name, we will first focus on the second part of the name, monohydrogensulfate which represents the anion. The name monohydrogensulfate ( $\text{HSO}_4^-$ ) comes from adding a proton ( $\text{H}^+$ ) to a sulfate cation ( $\text{SO}_4^{2-}$ ). Mind that protons ( $\text{H}^+$ ) are positively charged and therefore if we add a single  $\text{H}^+$  to a sulfate cation ( $\text{SO}_4^{2-}$ ) the charge will have to decrease a single unit, giving us  $\text{HSO}_4^-$ . As we can see, the name is directly related to the oxosalt anion and the number of hydrogens in the hydrosalt name. For example, phosphate ( $\text{PO}_4^{3-}$ ) is an oxosalt anion whereas hydrogenphosphate ( $\text{HPO}_4^{2-}$ ) and dihydrogenphosphate ( $\text{H}_2\text{PO}_4^-$ ) are both hydrosalt anions. An explanation about the charges: as phosphate has three negative charges, hydrogenphosphate has to have one less charge (that is  $2-$ ) and dihydrogenphosphate has to have two less negative charges (that is  $-1$ ). Some final hydrosalt anions examples:

|                        |                                  |
|------------------------|----------------------------------|
| carbonate              | $\text{CO}_3^{2-}$ (oxosalt ion) |
| monohydrogen carbonate | $\text{HCO}_3^-$ (hydrosalt ion) |

Above we saw how to name just the ending of the oxosalt with hydrogen anion. Now we can move forward to the whole naming of the salt. We just need to add the name of the element in the first place, and for example,  $\text{NaH}_2\text{BO}_3$  would be named sodium dihydrogenborate. If the first ion—the cation—is a transition metal cation (a type two cation) we need to include in parenthesis the valence of the cation. For example,  $\text{Fe}(\text{H}_2\text{BO}_3)_2$  would be named iron(II) dihydrogenborate. More examples:

|                               |                                    |
|-------------------------------|------------------------------------|
| sodium carbonate              | $\text{Na}_2\text{CO}_3$ (oxosalt) |
| sodium monohydrogen carbonate | $\text{NaHCO}_3$ (hydrosalt)       |



## Sample Problem 9

Name or formulate the following hydrosalts: (a) Magnesium hydrogensulfate (b) Sodium hydrogen carbonate (c)  $\text{LiHCO}_3$  (d)  $\text{Mg}(\text{H}_2\text{PO}_4)_2$

**SOLUTION**

The formula of Magnesium hydrogensulfate is  $\text{Mg}(\text{HSO}_4)_2$  as the formula for monohydrogen sulfate is  $\text{HSO}_4^-$  and the valence of magnesium is  $\text{Mg}^{2+}$ . The formula for Sodium monohydrogen carbonate is  $\text{NaHCO}_3$  as it results from combining  $\text{Na}^+$  and  $\text{HCO}_3^-$ . Mind monohydrogen carbonate results from adding a hydrogen ion  $\text{H}^+$  to a carbonate  $\text{CO}_3^{2-}$  ion. The name for  $\text{LiHCO}_3$  is lithium monohydrogen carbonate, whereas the name for  $\text{Mg}(\text{H}_2\text{PO}_4)_2$  is magnesium dihydrogenphosphate.

**STUDY CHECK**

Name or formulate the following hydrosalts: (a)  $\text{LiHS}_2\text{O}_3$  (b)  $\text{LiH}_2\text{PO}_4$  (c) sodium hydrogenphosphate

►Answer: (a) lithium monohydrogenthiosulfate (b) Lithium dihydrogenphosphate (c)  $\text{Na}_2\text{HPO}_3$

**Hydrates** Some chemicals contain water molecules trapped in their structure and therefore water molecules ( $\text{H}_2\text{O}$ ) are often indicated in chemical formulas. These types of chemicals containing water are called *hydrates*, precisely because hydrate means water. Examples of hydrates are  $\text{BeSO}_4 \cdot 4\text{H}_2\text{O}$  or  $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$  called respectively beryllium sulfate tetrahydrate and copper(II) sulfate pentahydrate. To formulate hydrates you just need to use prefixes such as mono, di, tetra—the same ones we use to name covalent chemicals, see Table ??—to indicate the number of water molecules in the chemical and end the name with *hydrate*. As a note, warming up hydrates (e.g.  $\text{BeSO}_4 \cdot 4\text{H}_2\text{O}$ ) results in the release of water producing a dehydrated or *anhydrous* compound (e.g.  $\text{BeSO}_4$ ). A final example of hydrate naming:

Sodium sulfate pentahydrate

 $\text{Na}_2\text{SO}_4 \cdot 5\text{H}_2\text{O}$  (a hydrate)

## Sample Problem 10

Name or formulate the following hydrates: (a) Nickel(II) permanganate dihydrate (b) Sodium nitrate monohydrate (c)  $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$  (d)  $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$

**SOLUTION**

The formula for Nickel(II) permanganate is  $\text{Ni}(\text{MnO}_4)_2$ , therefore the formula for Nickel(II) permanganate dihydrate is  $\text{Ni}(\text{MnO}_4)_2 \cdot 2\text{H}_2\text{O}$ . The formula for Sodium nitrate is  $\text{NaNO}_3$ , therefore  $\text{NaNO}_3 \cdot \text{H}_2\text{O}$  is Sodium nitrate monohydrate. The name for  $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$  is sodium carbonate decahydrate and  $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$  is magnesium sulfate heptahydrate.

**STUDY CHECK**

Name or formulate the following hydrates: (a)  $\text{LiNO}_3 \cdot \text{H}_2\text{O}$  (b)  $\text{Na}_3\text{PO}_4 \cdot 3\text{H}_2\text{O}$  (c) sodium sulfate tetrahydrate

►Answer: (a) lithium nitrate monohydrate (b) sodium phosphate trihydrate (c)  $\text{Na}_2\text{SO}_4 \cdot 4\text{H}_2\text{O}$

**Common naming** Some of the chemicals are normally referred to by a common name that does not involve the use of any chemical naming rules. An example would be  $\text{H}_2\text{O}$



normally referred to as water instead of its standard name which is dihydrogen oxide. You can find more names in Table ???. Another example:

|      |                                 |                          |
|------|---------------------------------|--------------------------|
| NaCl | Sodium chloride (standard name) | Table salt (common name) |
|------|---------------------------------|--------------------------|

| Table ??? List of common chemicals |                    |                                        |                  |
|------------------------------------|--------------------|----------------------------------------|------------------|
| Chemical                           | Name               | Chemical                               | Name             |
| H <sub>2</sub> O                   | Water              | Mg(OH) <sub>2</sub>                    | Milk of magnesia |
| NH <sub>3</sub>                    | Ammonia            | N <sub>2</sub> O                       | Laughing gas     |
| CH <sub>4</sub>                    | Methane            | CaCO <sub>3</sub>                      | Marble           |
| CO <sub>2</sub>                    | Dry ice            | CaO                                    | Quicklime        |
| NaCl                               | Table salt         | NaHCO <sub>3</sub>                     | Baking Soda      |
| NaHCO <sub>3</sub>                 | Sodium Bicarbonate | MgSO <sub>4</sub> · 7 H <sub>2</sub> O | Epsom Salt       |

#### Sample Problem 11

Name or formulate the following common chemicals: milk of magnesia and dry ice.

#### **SOLUTION**

The formula for milk of magnesia is Mg(OH)<sub>2</sub> (magnesium hydroxide), whereas dry ice is the common name for CO<sub>2</sub>, carbon dioxide.

#### **STUDY CHECK**

Name or formulate the following common chemicals: (a) ammonia (b) methane

►Answer: (a) NH<sub>3</sub> (nitrogen trihydride) (b) CH<sub>4</sub> (carbon tetrahydride)



## CHAPTER 3

### IONS & IONIC CHARGES

**3.1** Indicate if the following chemical species represent an atom, and anion or a cation: (a)  $\text{Fe}^{2+}$  (b)  $\text{Cl}^-$  (c) Ag

**3.2** Indicate if the following chemical species represent an atom, and anion or a cation: (a) Cs (b)  $\text{Cs}^+$  (c)  $\text{N}^{3-}$

**3.3** Identify the ionic state of the following elements. If needed, indicate the existence of multiple ionic states: (a) H (b) O (c) N (d) F (e) Mn

**3.4** Identify the ionic state of the following elements. If needed, indicate the existence of multiple ionic states: (a) Li (b) V (c) Cl (d) S (e) Cr (f) Sr (g) Ni

### COVALENT COMPOUNDS

**3.5** Name or formulate the following covalent compounds: (a) NO (b) Dichlorine monofluoride (c)  $\text{NO}_2$

**3.6** Name or formulate the following covalent compounds: (a) Chlorine Monofluoride (b)  $\text{N}_2\text{O}$  (c) Nitrogen trifluoride

**3.7** Name or formulate the following covalent compounds: (a)  $\text{SO}_3$  (b) Disulfur dichloride (c)  $\text{SO}_2$  (d) Disulfur tetrachloride

**3.8** Name or formulate the following covalent compounds: (a)  $\text{P}_4\text{S}_3$  (b) Sulfur Tetrafluoride (c)  $\text{As}_2\text{O}_5$  (d) Sulfur trioxide

### IONIC COMPOUNDS

**3.9** Classify the following chemicals in two groups, justifying your classification: (a) NaCl (b)  $\text{CO}_2$  (c)  $\text{FeCl}_3$  (d)  $\text{N}_2\text{O}_3$  (e)  $\text{SO}_3$  (f)  $\text{Ca}_3\text{N}_2$

**3.10** Combine the following ions: (a)  $\text{Na}^+ + \text{Cl}^-$  (b)  $\text{Na}^+ + \text{Se}^{2-}$  (c)  $\text{Na}^+ + \text{P}^{3-}$  (d)  $\text{Mg}^{2+} + \text{Cl}^-$  (e)  $\text{Mg}^{2+} + \text{O}^{2-}$  (f)  $\text{Mg}^{2+} + \text{N}^{3-}$

**3.11** Name or formulate the following ionic (Type I) compounds: (a) Magnesium iodide (b)  $\text{Ca}_3\text{P}_2$  (c) Lithium

nitride (d) MgF

**3.12** Name or formulate the following ionic (Type I) compounds: (a) Magnesium fluoride (b) CaS (c) Barium phosphide (d)  $\text{Mg}_3\text{N}_2$

**3.13** Name the following compounds: (a) NaCl (b)  $\text{Ca}_3\text{N}_2$  (c)  $\text{MgI}_2$  (d) SrS (e) RbCl (f) KF

**3.14** Combine the following ions: (a)  $\text{Cs}^+ + \text{F}^-$  (b)  $\text{Sr}^{2+} + \text{O}^{2-}$  (c)  $\text{Be}^{2+} + \text{C}^{4-}$  (d)  $\text{Li}^+ + \text{I}^-$

**3.15** Classify the following chemicals in two groups. Justify your classification. (a) NaCl (b)  $\text{MnO}_2$  (c)  $\text{FeCl}_3$  (d) SrO (e)  $\text{Li}_3\text{N}$  (f) NiO

**3.16** Formulate the following compounds: (a) Copper(I) oxide (b) Copper(II) nitride (c) Nickel(III) oxide (d) Manganese(IV) oxide

**3.17** Name the following compounds: (a) NiO (b)  $\text{Cr}_2\text{O}_3$  (c) VO (d)  $\text{MnO}_4$

**3.18** Formulate the following compounds: (a) Iron(II) nitride (b) Copper(I) sulfide (c) Chromium(III) iodide (d) Palladium(IV) phosphide (e) Manganese(VI) oxide

**3.19** Name the following compounds: (a)  $\text{Ni}_2\text{O}_3$  (b)  $\text{Fe}_3\text{N}_2$  (c)  $\text{Cr}_2\text{O}_3$  (d)  $\text{Ni}_3\text{P}_2$  (e)  $\text{Ru}_2\text{Se}_3$

**3.20** Name the following compounds: (a) FeO (b) CrN (c)  $\text{ZnI}_2$  (d) CoS (e)  $\text{MnF}_3$  (f)  $\text{Cu}_2\text{C}$  (g)  $\text{Ag}_2\text{O}$

**3.21** Name or formulate the following ionic (Type II) compounds: (a)  $\text{Fe}_3\text{P}_2$  (b) Copper(II) iodide (c)  $\text{Fe}_3\text{N}_2$  (d) Iron(II) sulfide

**3.22** Name or formulate the following ionic (Type II) compounds: (a)  $\text{Fe}_2\text{S}_3$  (b) Gold(III) chloride (c) FeO (d) Vanadium(V) nitride



**3.23** Name or formulate the following ionic (Type II) compounds: (a)  $\text{FeI}_2$  (b) Lead(IV) sulfide (c)  $\text{FeBr}_2$

**3.24** Name or formulate the following ionic (Type II) compounds: (a) Manganese(IV) oxide (b)  $\text{FeCl}_2$  (c) Copper(I) oxide

#### ACIDS AND HYDROXIDES

**3.25** Name or formulate the following acids or bases: (a)  $\text{HCl}$  (b) Hydrofluoric Acid (c)  $\text{Mg}(\text{OH})_2$

**3.26** Name or formulate the following acids or bases: (a) Sulfuric Acid (b)  $\text{H}_2\text{CO}_3$  (c) Lithium hydroxide

**3.27** From the following chemicals identify acids and bases: (a)  $\text{KOH}$  (b)  $\text{LiOH}$  (c)  $\text{CH}_3\text{OH}$

**3.28** From the following chemicals identify acids and bases: (a)  $\text{H}_2\text{SO}_3$  (b)  $\text{NH}_3$  (c)  $\text{Ca}(\text{OH})_2$

**3.29** From the following chemicals identify hydric acids and oxoacids: (a)  $\text{HF}$  (b)  $\text{H}_2\text{SO}_3$  (c)  $\text{H}_2\text{S}$

**3.30** From the following chemicals identify hydric acids and oxoacids: (a)  $\text{H}_3\text{BO}_3$  (b)  $\text{HCl}$  (c)  $\text{HI}$

**3.31** Identify the redox number of the central atom of the following oxoacids: (a)  $\text{H}_2\text{CrO}_4$  (b)  $\text{H}_2\text{Cr}_2\text{O}_7$  (c)  $\text{HMnO}_4$

**3.32** Identify the redox number of the central atom of the following oxoacids: (a)  $\text{H}_2\text{MnO}_4$  (b)  $\text{HReO}_3$  (c)  $\text{H}_2\text{SiO}_3$

**3.33** Identify the most oxidized acid: (a)  $\text{H}_3\text{AsO}_4$  or  $\text{H}_3\text{AsO}_3$  (b)  $\text{H}_2\text{XeO}_4$  or  $\text{H}_4\text{XeO}_6$

**3.34** Identify the most reduced acid: (a)  $\text{H}_2\text{RuO}_4$  or  $\text{HRuO}_4$  (b)  $\text{HTcO}_4$  or  $\text{H}_2\text{TcO}_4$

**3.35** Identify the most oxidant acid: (a)  $\text{H}_2\text{CrO}_4$  or  $\text{H}_2\text{Cr}_2\text{O}_7$  (b)  $\text{HNO}_3$  or  $\text{HNO}_2$

**3.36** Identify the most oxidant acid: (a)  $\text{H}_2\text{S}_2\text{O}_6$  or  $\text{H}_2\text{SO}_4$  (b)  $\text{H}_2\text{SeO}_4$  or  $\text{H}_2\text{SeO}_3$

#### NAMING OF COMPLEX SALTS AND COMMON CHEMICALS

**3.37** Name or formulate the following oxoanions: (a)  $\text{ClO}_4^-$  (b)  $\text{PO}_4^{3-}$  (c)  $\text{SO}_4^{2-}$  (d)  $\text{CO}_3^{2-}$  (e)  $\text{NO}_3^-$  (f)  $\text{CrO}_4^{2-}$  (g)  $\text{BO}_3^{3-}$

**3.38** Name or formulate the following (Type I) oxosalts: (a)  $\text{Mg}(\text{NO}_3)_2$  (b) Sodium permanganate (c)  $\text{KMnO}_4$  (d) Calcium carbonate (e)  $\text{Li}_3\text{PO}_4$

**3.39** Name or formulate the following (Type I) oxosalts: (a) Lithium sulfate (b)  $\text{Na}_2\text{CrO}_4$  (c) Lithium sulfite (d)  $\text{Cs}_2\text{Cr}_2\text{O}_7$  (e) Calcium sulfate

**3.40** Name or formulate the following compounds: (a)  $\text{Na}_2\text{SO}_4$  (b)  $\text{KNO}_3$  (c)  $\text{CaCO}_3$  (d)  $\text{Ca}(\text{NO}_2)_2$  (e)  $\text{SrSO}_3$

**3.41** Combine the following ions: (a)  $\text{Na}^+ + \text{PO}_4^{3-}$  (b)  $\text{Li}^+ + \text{MnO}_4^-$  (c)  $\text{Mg}^{2+} + \text{NO}_3^-$  (d)  $\text{Ca}^{2+} + \text{CO}_3^{2-}$  (e)  $\text{Cs}^+ + \text{Cr}_2\text{O}_7^{2-}$  (f)  $\text{K}^+ + \text{BO}_3^{3-}$

**3.42** Combine the following polyatomic ions: (a)  $\text{Na}^+ + \text{NO}_3^-$  (b)  $\text{Na}^+ + \text{CO}_3^{2-}$  (c)  $\text{Na}^+ + \text{PO}_4^{3-}$  (d)  $\text{Ca}^{2+} + \text{CO}_3^{2-}$  (e)  $\text{Ca}^{2+} + \text{PO}_4^{3-}$

**3.43** Name or formulate the following (Type II) oxosalts: (a)  $\text{Cr}_2(\text{SO}_4)_3$  (b) zinc(II) carbonate (c)  $\text{Fe}(\text{MnO}_4)_3$

**3.44** Name or formulate the following (Type II) oxosalts: (a) cobalt(III) carbonate (b)  $\text{Fe}(\text{ClO}_4)_3$  (c) zinc(II) carbonate

**3.45** Name or formulate the following hydrosalts: (a)  $\text{NaHCO}_3$  (b) Calcium Hydrogencarbonate (c)  $\text{Al}(\text{HSO}_4)_3$

**3.46** Name or formulate the following hydrosalts: (a) Sodium dihydrogenphosphate (b)  $\text{LiH}_2\text{PO}_4$  (c) Silver monohydrogenphosphate

**3.47** Name or formulate the following hydrates: (a)  $\text{Al}_2(\text{SO}_4)_3 \cdot 3\text{H}_2\text{O}$  (b) Silver phosphate dihydrate



**3.48** Name or formulate the following hydrates:  
(a)  $\text{KMnO}_4 \cdot 4 \text{H}_2\text{O}$  (b) Lithium sulfate tetrahydrate

**3.49** Name or formulate the following compounds:  
(a)  $\text{MgSO}_4$  (b)  $\text{Ni}(\text{SO}_4)_3$  (c) Cobalt(II) nitrate  
(d) Cobalt(II) sulfate dihydrate (e)  $\text{KHCO}_3$

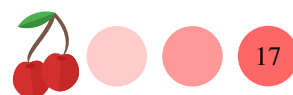
**3.50** Name or formulate the following compounds:  
(a)  $\text{Ca}(\text{NO}_3)_2$  (b)  $\text{Ca}(\text{HCO}_3)_2$  (c) Nickel(II) sulfate  
(d) Nickel(II) sulfate tetrahydrate (e)  $\text{NaH}_2\text{PO}_4$

**3.51** Name or formulate the following compounds:  
(a)  $\text{MnSO}_4$  (b)  $\text{CuNO}_3$  (c)  $\text{Cr}_2(\text{CO}_3)_3$  (d)  $\text{V}(\text{NO}_2)_2$   
(e)  $\text{FeSO}_3$

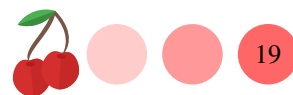
**3.52** Name or formulate the following pairs or ions:  
(a) carbonate and monohydrogencarbonate (b) sulfate and monohydrogensulfate (c) chromate and monohydrogenschromate (d) phosphate and dihydrogenphosphate (e) phosphate and monohydrogenphosphate (f) borate and dihydrogenphosborate













**Answers 3.1** (a) cation (b) anion (c) atom **3.2** (a) atom (b) cation (c) anion **3.3** (a)  $\text{H}^+$  (b) (e) Mn (multiple) **3.4** (a) Li  $\text{Li}^+$  (b) V (multiple) (c) Cl  $\text{Cl}^-$  (d) S  $\text{S}^{2-}$  (e) Cr (multiple) (f) Sr Sr **3.5** (a) nitrogen monoxide (b)  $\text{Cl}_2\text{F}$  (c) nitrogen dioxide **3.6** (a)  $\text{ClF}$  (b) dinitrogen monoxide (c) trioxide (b)  $\text{S}_2\text{Cl}$  (c) sulfur dioxide (d)  $\text{S}_2\text{Cl}_4$  **3.8** (a) tetraphosphorus trisulfide (b)  $\text{SF}_4$  (c) diarsenic **3.9** (a) NaCl (group ionic) (b)  $\text{CO}_2$  (group covalent) (c)  $\text{FeCl}_3$  (group ionic) (d)  $\text{N}_2\text{O}_3$  (group covalent) (f)  $\text{Ca}_3\text{N}_2$  (group ionic) **3.10** (a)  $\text{Na}^+ + \text{Cl}^-$  (NaCl) (b)  $\text{Na}^+ + \text{Se}^{2-}$  ( $\text{Na}_2\text{Se}$ ) (c)  $\text{Na}^+ + \text{P}^{3-}$  ( $\text{Na}_3\text{P}$ ) (d)  $\text{Mg}^{2+} + \text{O}^{2-}$  (MgO) (e)  $\text{Mg}^{2+} + \text{N}^{3-}$  ( $\text{Mg}_3\text{N}_2$ ) **3.11** (a) Magnesium iodide ( $\text{MgI}_2$ ) (b) calcium phosphide (c) Lithium nitride ( $\text{Li}_3\text{N}$ ) (d) MgF (Magnesium fluoride) **3.12** (a) Magnesium fluoride (b) calcium sulfide (c) Barium phosphide ( $\text{Ba}_3\text{P}_2$ ) (d)  $\text{Mg}_3\text{N}_2$  (magnesium nitride) **3.13** (a) NaCl (sodium chloride) (b)  $\text{Ca}_3\text{N}_2$  (calcium nitride) (c)  $\text{MgI}_2$  (magnesium iodide) (d) SrS (strontium sulfide) (e) RbCl (rubidium chloride) (f)  $\text{BaF}_2$  (barium fluoride) **3.14** (a) (CsF) (b) (SrO) (c) ( $\text{Be}_2\text{C}$ ) (d) (LiI) **3.15** (a) NaCl (group ionic type I) (b) MnCl<sub>2</sub> (group ionic type II) (c)  $\text{FeCl}_3$  (group ionic type II) (d) SrO (group ionic type I) (e)  $\text{Li}_3\text{N}$  (group ionic type I) (f) NiO (group ionic type II) **3.16** (a) Copper(I) oxide ( $\text{Cu}_2\text{O}$ ) (b) Copper(II) nitride ( $\text{Cu}_3\text{N}_2$ ) (c) Nickel(III) oxide ( $\text{Ni}_2\text{O}_3$ ) (d) Manganese(II) oxide ( $\text{MnO}$ ) (e) Vanadium(V) oxide ( $\text{V}_2\text{O}_5$ ) (f) Chromium(VI) oxide ( $\text{CrO}_3$ ) (g)  $\text{Cr}_2\text{O}_3$  (chromium(III) oxide) (h)  $\text{Mn}_2\text{O}_7$  (manganese(VII) oxide) **3.17** (a) NiO (Nickel(II) oxide) (b)  $\text{Cr}_2\text{O}_3$  (Chromium(III) oxide) (c) VO (Vanadium(II) oxide) (d)  $\text{CrO}_2$  (Chromium(IV) oxide) (e)  $\text{Mn}_2\text{O}_7$  (manganese(VII) oxide) **3.18** (a) Iron(II) nitride ( $\text{Fe}_3\text{N}_2$ ) (b) Copper(I) sulfide ( $\text{Cu}_2\text{S}$ ) (c) Chromium(III) iodide ( $\text{CrI}_3$ ) (d) Radium(IV) phosphide ( $\text{Pd}_3\text{P}_4$ ) (e) Manganese(VI) oxide ( $\text{MnO}_3$ ) **3.19** (a)  $\text{Ni}_2\text{O}_3$  (nickel(III) oxide) (b)  $\text{Fe}_3\text{N}_2$  (iron(II) nitride) (c)  $\text{Cr}_2\text{O}_3$  (chromium(III) oxide) (d)  $\text{Ni}_3\text{P}_2$  (nickel(II) phosphide) (e)  $\text{Ru}_2\text{Se}_3$  (ruthenium(III) selenide) (f)  $\text{Co}_2\text{O}_3$  (cobalt(III) oxide) (g)  $\text{Fe}_2\text{O}_3$  (iron(III) oxide) (h)  $\text{CrN}$  (Chromium(III) nitride) (i)  $\text{ZnI}_2$  (Zinc (II) iodide) (j)  $\text{CoS}$  (Cobalt(II) sulfide) (k)  $\text{MnF}_2$  (manganese(II) fluoride) (l)  $\text{Cu}_2\text{C}$  (copper(IV) carbide) (m)  $\text{Ag}_2\text{O}$  (silver oxide) **3.21** (a)  $\text{Fe}_3\text{P}_2$  (Iron(III) phosphide) (b)  $\text{CuI}_2$  (Copper(II) iodide) (c)  $\text{Fe}_3\text{N}_2$  (Iron(II) nitride) (d)  $\text{FeS}$  (Iron(II) sulfide) **3.22** (a)  $\text{Fe}_2\text{S}_3$  (Iron(III) sulfide) (b)  $\text{AuCl}_3$  (Gold(III) chloride) (c)  $\text{FeO}$  (Iron(II) oxide) (d)  $\text{V}_3\text{N}_5$  (Vanadium(V) nitride) **3.23** (a)  $\text{FeI}_2$  (Iron(II) iodide) (b)  $\text{PbS}_2$  (Lead(II) sulfide) (c)  $\text{FeBr}_2$  (Iron(II) bromide) **3.24** (a) Manganese(IV) oxide ( $\text{MnO}_2$ ) (b)  $\text{FeCl}_2$  (Iron(II) chloride) (c)  $\text{Cu}_2\text{O}$  (Copper(I) oxide) **3.25** (a) HCl (hydrochloric acid) (b) Hydrofluoric Acid (HF) (c)  $\text{Mg}(\text{OH})_2$  (magnesium hydroxide) (d)  $\text{H}_2\text{SO}_4$  (sulfuric acid) **3.26** (a) Sulfuric Acid ( $\text{H}_2\text{SO}_4$ ) (b)  $\text{H}_2\text{CO}_3$  (carbonic acid) (c) Lithium hydroxide ( $\text{LiOH}$ ) **3.27** (a) KOH (potassium hydroxide) (b)  $\text{CH}_3\text{OH}$  (acid, organic) **3.28** (a)  $\text{H}_2\text{SO}_3$  (acid, inorganic) (b)  $\text{NH}_3$  (base) (c)  $\text{Ca}(\text{OH})_2$  (calcium hydroxide) (d)  $\text{H}_2\text{SO}_3$  (oxacid) (e)  $\text{H}_2\text{S}$  (hydracid) **3.30** (a)  $\text{H}_3\text{BO}_3$  (oxacid) (b) HCl (hydracid) **3.31** (a)  $\text{H}_2\text{CrO}_4$  redox=6 (b)  $\text{H}_2\text{Cr}_2\text{O}_7$  redox=6 (c)  $\text{HMnO}_4$  redox=7 **3.32** (a)  $\text{H}_2\text{MnO}_4$  redox=6 (b)  $\text{H}_2\text{SiO}_3$  redox=4 **3.33** (a)  $\text{H}_3\text{As}^{\text{V}}\text{O}_4$  (b)  $\text{H}_4\text{Xe}^{\text{VIII}}\text{O}_6$  **3.34** (a)  $\text{H}_2\text{Ru}^{\text{VI}}\text{O}_4$  (b)  $\text{HTc}^{\text{VII}}\text{O}_4$  **3.35** (a)  $\text{H}_2\text{S}_2\text{O}_6$  (b)  $\text{H}_2\text{SeO}_3$  **3.37** (a) perchlorate (b) phosphate (c) sulfate (d) carbonate (e) nitrate (f) nitrite **3.38** (a) magnesium nitrate (b)  $\text{NaMnO}_4$  (c) potassium permanganate (d)  $\text{CaCO}_3$  (e) lithium phosphate (f) sodium dichromate (g)  $\text{Li}_2\text{SO}_3$  (d) caesium dichromate (e)  $\text{CaSO}_4$  **3.40** (a) sodium sulfate (b) calcium carbonate (c) calcium nitrite (e) strontium sulfite **3.41** (a)  $\text{Na}_3\text{PO}_4$  (b)  $\text{LiMnO}_4$  (c)  $\text{Mg}_3\text{N}_2$  (d)  $\text{Cs}_2\text{Cr}_2\text{O}_7$  (f)  $\text{K}_3\text{BO}_3$  **3.42** (a)  $\text{NaNO}_3$  (b)  $\text{Na}_2\text{CO}_3$  (c)  $\text{Na}_3\text{PO}_4$  (d)  $\text{CaCO}_3$  (e)  $\text{Ca}_3(\text{PO}_4)_2$  **3.43** (a)  $\text{ZnSO}_4$  (b)  $\text{ZnCO}_3$  (c) iron(III) permanganate **3.44** (a)  $\text{Co}_2(\text{CO}_3)_2$  (b) iron(III) perchlorate (c)  $\text{ZnCO}_3$  (d) hydrogen carbonate (b)  $\text{Ca}(\text{HCO}_3)_2$  (c) aluminum hydrogensulfate **3.46** (a)  $\text{NaH}_2\text{PO}_4$  (b) lithium dihydrogenphosphate (c)  $\text{Ag}_2\text{HPO}_4$  **3.47** (a) aluminum phosphate trihydrate (b)  $\text{Ag}_3\text{PO}_4 \cdot 2\text{H}_2\text{O}$  **3.48** (a) potassium permanganate (b)  $\text{Li}_2\text{SO}_4 \cdot 4\text{H}_2\text{O}$  **3.49** (a) magnesium phosphate (b) nickel(III) sulfate (c)  $\text{Co}(\text{NO}_3)_2$  (d)  $\text{CoSO}_4 \cdot 2\text{H}_2\text{O}$  (e) monohydrogencarbonate **3.50** (a) calcium nitrate (b) Calcium monohydrogencarbonate (c)  $\text{NiSO}_4$  (d) sodium dihydrogenphosphate **3.51** (a) manganese(II) sulfate (b) copper(I) nitrate (c) chromium(III) nitrate (d)  $\text{Fe}(\text{NO}_3)_3$  (e) Iron(II) sulfite **3.52** (a)  $\text{CO}_3^{2-}$  and  $\text{HCO}_3^-$  (b)  $\text{SO}_4^{2-}$  and  $\text{HSO}_4^-$  (c)  $\text{CrO}_4^{2-}$  and  $\text{HCrO}_4^-$  (d)  $\text{PO}_4^{3-}$  and  $\text{HPO}_4^{2-}$  (e)  $\text{PO}_4^{3-}$  and  $\text{HPO}_4^{2-}$  (f)  $\text{BO}_3^{3-}$  and  $\text{H}_2\text{BO}_3^-$